

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Semestertoets 2

Kopiereg voorbehou

Semester Test 2

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Module EBN111
Elektrisiteit en Elektronika
25 April 2012

Module EBN111
Electricity and Electronics
25 April 2012

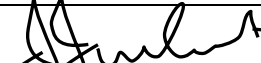


UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
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Eksamensinligting: <i>Exam information:</i>		MEMO Q2/Q3	
Maksimum punte: <i>Maximum marks:</i>	45	Volpunte: <i>Full marks:</i>	45
Duur van vraestel: <i>Duration of paper:</i>	90 minute <i>90 minutes</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total numer of pages (including this page):</i>		13	

BELANGRIK- IMPORTANT

1. Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
2. Alle vrae moet op die ANTWOORDBLAD beantwoord word.
All questions must be answered on the ANSWER SHEET.
3. Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
Students may do their own rough calculations on the question paper – the question paper will not be taken in.
4. Antwoorde moet eenhede insluit!
Answers must include units!
5. Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.
Final answers must be written as real numbers, and not as fractions.

Dosent(e): <i>Lecturer(s):</i>	1 J Joubert		
	2 JD le Roux		
Groephoof: Ek bevestig dat die vraestel die uitkomstes toets soos gespesifiseer in die studiehandleiding. <i>Group head: I confirm that the question paper evaluates the outcomes as specified in the study guide.</i>			
Groephoof: <i>Group Head:</i>	Prof J Joubert	Handtekening: <i>Signature:</i>	

VRAAG/QUESTION 2 [6]

Vraag 2.1 [4]

Gegee: $\beta = 99$ en $V_{BE} = 0.7$ V.

a) Indien $V_S = 5.7$ V, bepaal I_C . [2]

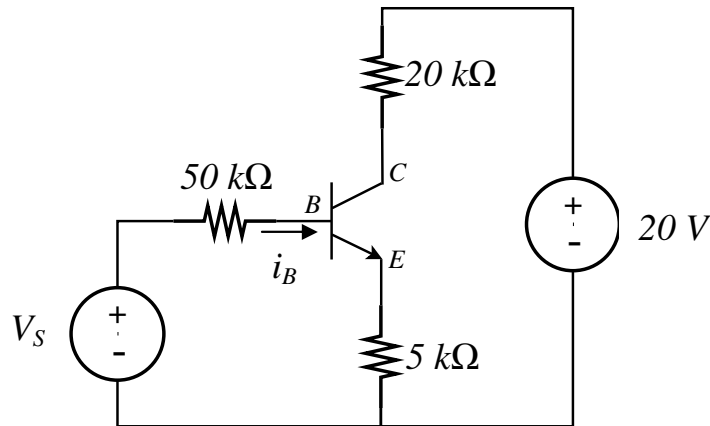
b) Indien $I_B = 5$ μ A, bepaal V_{CE} . [2]

Question 2.1 [4]

Given: $\beta = 99$ and $V_{BE} = 0.7$ V.

a) If $V_S = 5.7$ V, determine I_C . [2]

b) If $I_B = 5$ μ A, determine V_{CE} . [2]



a)

KVL:

$$-5.7 + 50k * i_B + 5k * i_E + 0.7 = 0$$

But:

$$\begin{aligned} i_E &= i_B + i_C \\ &= i_B + 99i_B \\ &= 100 i_B \end{aligned}$$

Thus:

$$(50k + 5k * 100) i_B = 5$$

$$i_B = 5/(550k)$$

$$i_B = 9.0909 \mu A$$

$$\text{Thus } i_C = 99 * 9.0909 \mu A = \mathbf{0.9 \text{ mA}}$$

b)

KVL:

$$-20 + 20k * i_C + V_{CE} + 5k * i_E = 0$$

$$20 = 20k * 99 i_B + V_{CE} + 5k * 100i_B$$

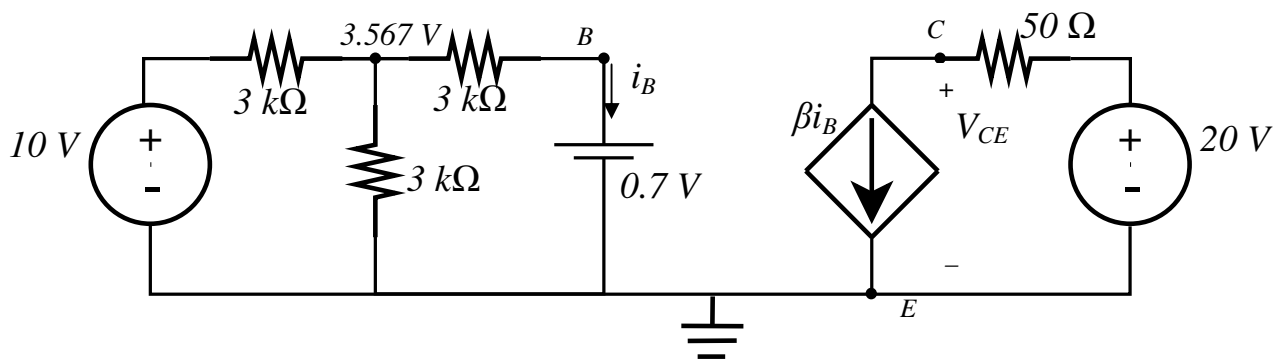
$$20 = (20k * 99 + 500k)i_B + V_{CE}$$

$$20 = (1980k + 500k) i_B + V_{CE}$$

$$20 = 2480k (5\mu) + V_{CE}$$

$$20 = 12.4 \text{ V} + V_{CE}$$

$$\mathbf{V_{CE} = 7.6 \text{ V}}$$

Vraag 2.2 [2]Bepaal V_{CE} indien $\beta = 100$.**Question 2.2 [2]**Determine V_{CE} if $\beta = 100$.

$$i_B = (3.567 - 0.7) / 3k = 2.867 / 3000 = 0.956 \text{ mA}$$

Thus:

$$-20 + \beta i_B * 50 + V_{CE} = 0$$

$$-20 + 100 * 0.956 \text{ mA} * 50 + V_{CE} = 0$$

$$-20 + 4.78 + V_{CE} = 0$$

$$V_{CE} = 15.22 \text{ V}$$

VRAAG/QUESTION 3 [16]

Vraag 3.1 [4]

Skryf V_0 in terme van R_F , R_I , en R_2 indien:

a) $V_I = 0$ V en $V_2 = 1$ V [2]

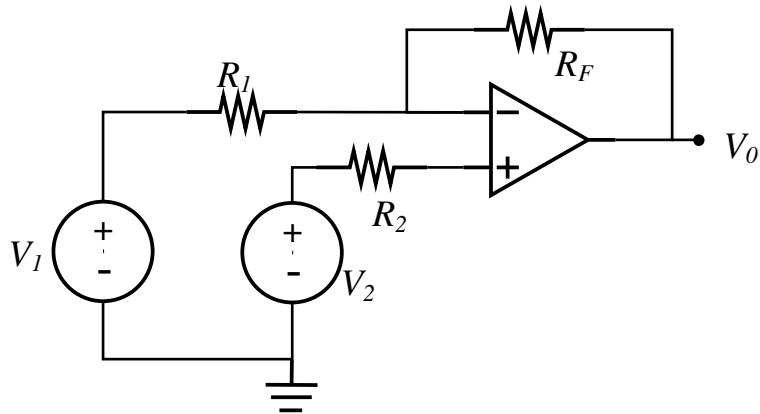
b) $V_2 = 0$ V en $V_I = 1$ V [2]

Question 3.1 [4]

Write V_0 in terms of R_F , R_I , and R_2 if:

a) $V_I = 0$ V and $V_2 = 1$ V [2]

b) $V_2 = 0$ V and $V_I = 1$ V [2]



a) Non-inverting amplifier

$$V_0 = (1 + R_F/R_I)$$

b) Inverting amplifier

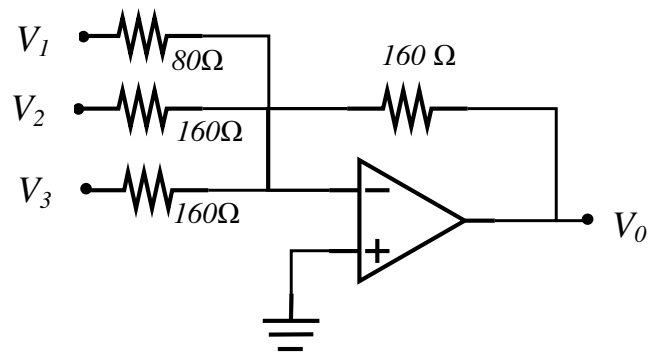
$$V_0 = -R_F/R_I$$

Vraag 3.2 [2]

Indien $V_0 = 60 \text{ V}$, $V_1 = 10 \text{ V}$ en $V_2 = 20 \text{ V}$, bepaal V_3 .

Question 3.2 [2]

If $V_0 = 60 \text{ V}$, $V_1 = 10 \text{ V}$ and $V_2 = 20 \text{ V}$, determine V_3 .



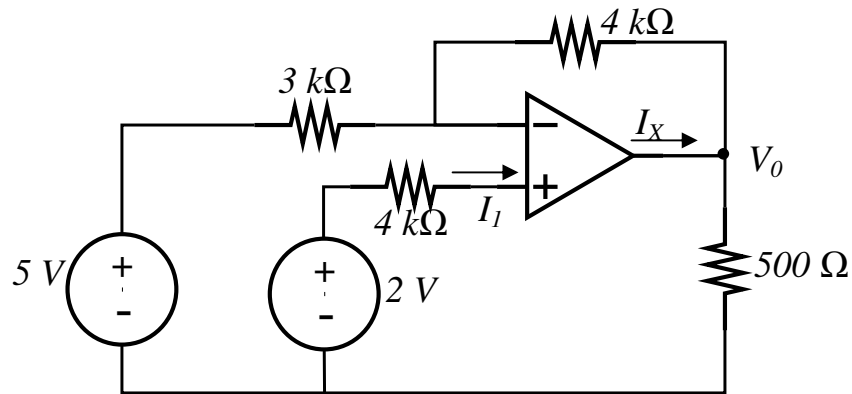
$$V_0 = -160 * (1/80 * V_1 + 1/160 * V_2 + 1/160 * V_3)$$

$$60 = - (2 * V_1 + V_2 + V_3)$$

$$60 = - (2 * 10 + 20 + V_3)$$

$$V_3 = - 60 - 40$$

$$V_3 = - 100 \text{ V}$$

Vraag 3.3 [3]Bepaal I_X , I_I en V_0 .**Question 3.3 [3]**Determine I_X , I_I and V_0 .

$$I_I = 0 \text{ A}$$

$$V_- = V_+ = 2 \text{ V}$$

KCL at node V_-

$$(5-2)/3\text{k} = (2 - V_0)/4\text{k}$$

$$1 = 0.5 - V_0/4$$

$$-0.5 = V_0/4$$

$$V_0 = -2 \text{ V}$$

$$I_X = V_0/500 + (V_0-2)/4\text{k}$$

$$I_X = -2/500 + (-2 - 2)/4\text{k}$$

$$I_X = -2/500 + (-4)/4\text{k}$$

$$I_X = -4\text{mA} - 1\text{mA}$$

$$I_X = -5 \text{ mA}$$

Vraag 3.4 [4]

Gebruik node-analise om twee vergelykings van die volgende vorm op te stel vir die stroombaan:

$$aV_1 + bV_2 = 48$$

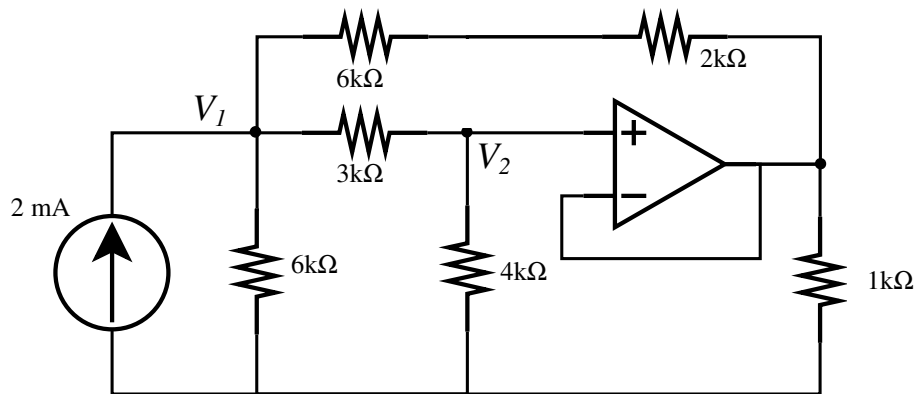
$$-4V_1 + cV_2 = d$$

Question 3.4 [4]

Use node analysis to set up two equations of the following form for the circuit:

$$aV_1 + bV_2 = 48$$

$$-4V_1 + cV_2 = d$$

**Node 1:**

$$2\text{mA} = V_1/6\text{k} + (V_1 - V_2)/3\text{k} + (V_1 - V_2)/8\text{k}$$

$$2 = V_1 (1/6 + 1/3 + 1/8) - V_2 (1/3 + 1/8)$$

$$2 = V_1 (5/8) - V_2 (11/24)$$

$$48 = 15 V_1 - 11 V_2$$

Amplifier is a buffer. Thus output is equal to V_2 .

Node 2:

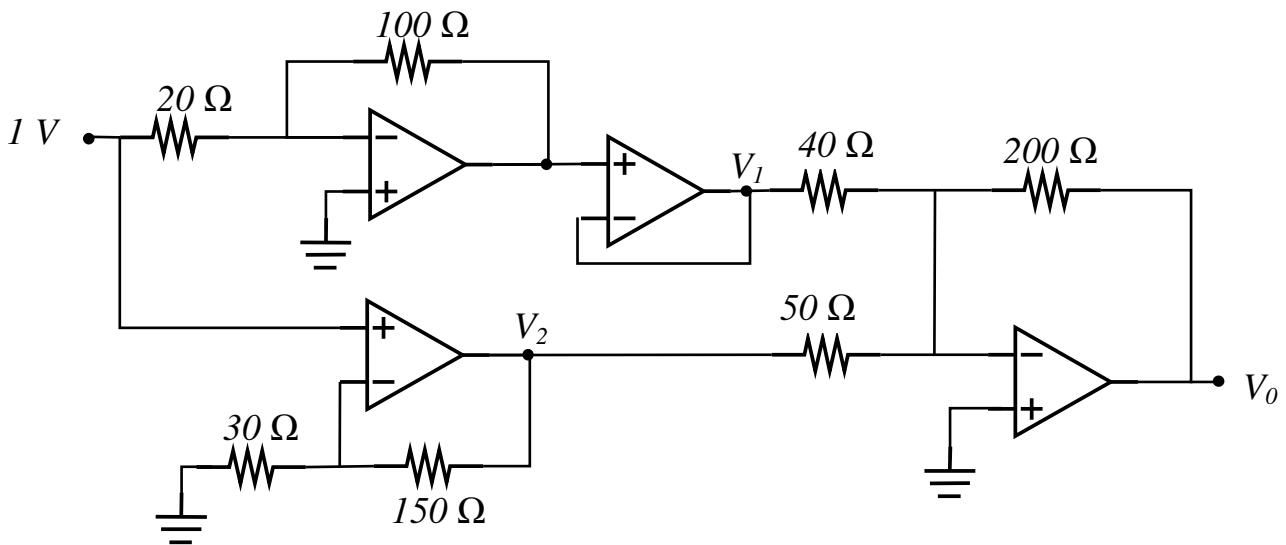
$$V_2 / 4\text{k} + (V_2 - V_1) / 3\text{k} = 0$$

$$V_2 (1/4 + 1/3) - V_1 / 3 = 0$$

$$7/12 V_2 - 1/3 V_1 = 0$$

$$7 V_2 - 4 V_1 = 0$$

$$-4 V_1 + 7 V_2 = 0$$

Vraag 3.5 [3]Vir die kaskade kringbaan, bepaal V_1 , V_2 en V_0 .**Question 3.5 [3]**For the cascade circuit, determine V_1 , V_2 and V_0 .

Inverting amplifier

$$V_1 = 1V * (-100/20)$$

$$= -5V$$

Non-inverting amplifier

$$V_2 = 1V * (1 + 150/30)$$

$$= 6V$$

Summing Amplifier

$$V_0 = -200 (1/40 * (-5) + 1/50 * 6)$$

$$= -5 * (-5) - 4 * 6$$

$$= 25 - 24$$

$$= 1V$$

VRAAG 2 – QUESTION 2 (6)

2.1	$I_C = 0.9 \text{ mA} \quad \checkmark \checkmark$	$V_{CE} = 7.6 \text{ V} \quad \checkmark \checkmark$
2.2	$V_{CE} = 15.22 \text{ V} \quad \checkmark \checkmark$	

VRAAG 3 – QUESTION 2 (16)

3.1	$a) V_0 = (1 + R_F/R_I) \quad \checkmark \checkmark$	
	$b) V_0 = -R_F/R_I \quad \checkmark \checkmark$	
3.2	$V_3 = -100 \text{ V} \quad \checkmark \checkmark$	
3.3	$I_I = 0 \text{ A} \quad \checkmark$	
	$I_X = -5 \text{ mA} \quad \checkmark$	
	$V_0 = -2 \text{ V} \quad \checkmark$	
3.4	$a = 15 \quad \checkmark$	$b = -11 \quad \checkmark$
	$c = 7 \quad \checkmark$	$d = 0 \quad \checkmark$
3.5	$V_I = -5 \text{ V} \quad \checkmark$	
	$V_2 = 6 \text{ V} \quad \checkmark$	
	$V_0 = 1 \text{ V} \quad \checkmark$	